

ED Analytics

DAMO-6994 Capstone

Analysis Pipeline

14%

About Project

Prep & Quality Engine

Dataset Explorer

Hypothesis Testing & Statist...

Executive Dashboard

Strategic Insights

Reports & Export

Enterprise Workspace

CIHI NACRS Analytics

< Start

Stage 1/7 · About Project

CREDENTIALS VERIFIED

EXECUTIVE RESEARCH OVERVIEW · CAPSTONE PLATFORM

Canadian Emergency Department Analytics Platform

Explanatory biostatistical modeling of ED Length of Stay (LOS) and longitudinal hospital resource burden using aggregate CIHI NACRS administrative data (2003–2021).

Program: Master of Data Analytics

Source: CIHI NACRS Aggregate

Time Span: 19 Fiscal Years

Quality Audit: 100% Validated

CORE RESEARCH THESIS

What systemic and clinical factors drive ED stay duration and cumulative emergency strain?

Synthesizes 8,685+ pre-aggregated records across 175.8M visits to isolate independent predictors of operational delays, evaluate patient demographic disparities, and forecast near-term demand trajectories.

CLINICAL RECORDS

8,685+

6 Harmonized Worksheets

TIME HORIZON

19 Years

2003–04 to 2021–22

HYPOTHESES

5 Tested

H1–H5 Verified (p < .0001)

DATA INTEGRITY

100%

0 Missing / 0 Duplicates

Explore Data Pipeline

METHODOLOGICAL FRAMEWORK

Four Foundations of the Analytics Platform

01 · SOURCE DATA

Multi-Cohort Integration

Combines 19-year longitudinal trends, triage presentations, presenting complaints, and demographic breakdowns from CIHI NACRS.

02 · LONGITUDINAL

19-Year Trend Analysis

Tracks annual patient volume, median stay durations, pandemic shifts, and cumulative Total ED-Minutes (TEM) system burden.

03 · INFERENCE

Statistical Testing

Rigorous frequency-weighted non-parametric tests (Kruskal-Wallis, Mann-Whitney U), Dunn post-hoc contrasts, and WLS regression models.

04 · GOVERNANCE

Privacy-Safe Research

100% aggregate administrative health data. Non-identifiable group statistics strictly for health system operations research.

SYSTEMIC CHALLENGE

Where Does ED Bottlenecking Occur?

Canadian emergency departments navigate multifaceted throughput constraints across aging populations, acuity variations, and hospital bed shortages. Length of Stay (LOS) serves as the primary macro-indicator of system strain.

Acuity Gradient

Resuscitation and Emergent presentations require intensive stabilization, extending median stay durations.

Geriatric Burden

Older adults (65+) experience complex multimorbidity, driving higher aggregate median stay times.

Admission Delays

Admitted visits exhibit severe boarding delays (10.60h median vs. 2.50h for discharged patients).

TEM Volume Strain

Total ED-Minutes = Visit Volume × Median Stay Duration captures compounding system load.

Focus: NACRS Ambulatory Cohorts

SYSTEM-LEVEL ANALYSIS

ANALYTICAL FRAMEWORK

Five Core Research Objectives

01 → 05 Pipeline

01. Explain Stay Drivers

Quantify how triage acuity, age groups, sex, and visit outcomes drive variation in reported ED length of stay.

02. Measure TEM Burden

Compute Total ED-Minutes (TEM) = Visit Volume × Median LOS to evaluate cumulative longitudinal resource strain.

03. WLS Regression

Execute weighted least squares regressions to isolate the independent effect of urgency scores on stay duration.

04. High-Burden Cohorts

Identify combinations of triage level, age segment, and clinical complaints creating disproportionate bed occupancy.

05. Longitudinal Forecasting

Analyze 19-year longitudinal trajectories and forecast near-term demand using Simple Exponential Smoothing (SES) with 95% confidence bands.

ANALYTICAL VERIFICATION

Scientific Rigor, Data Pipeline & Architecture

5 Hypotheses (H1-H5)7-Stage PipelineTech Stack

H1

CTAS Triage Acuity & LOS

"Does median ED stay differ across CTAS urgency levels (1–5)?"

CTAS I Resuscitation (4.60h / 276m) and CTAS II Emergent (4.80h / 288m) exhibit substantially longer stays than CTAS V Non-Urgent (1.33h / 80m) ($\epsilon^2 = 0.7251$).

Method: Weighted Kruskal-Wallis & Dunn Post-Hoc

H2

Admission Status & LOS

"Does length of stay differ between admitted and non-admitted visits?"

Admitted visits follow significantly longer aggregate stay durations (10.60h median stay vs. 2.50h for non-admitted visits; $r_b = 0.9981$).

Method: Weighted Mann-Whitney U Test

H3

WLS Regression Predictors of LOS

"Does CTAS urgency score predict reported median stay duration?"

Urgency score significantly predicts reported stay duration ($\beta_i = -73.92$ min/unit / -1.23 h/unit, $R^2 = 0.6256$, $N = 174.2M$ visits).

Method: Weighted Least Squares (WLS) Regression

H4

Patient Age Group & LOS

"Does reported median stay differ across broad age categories?"

Older Adults 65+ (4.17h / 250m) experience longer aggregate stays than pediatric patients (2.05h / 123m) ($\epsilon^2 = 0.7218$).

Method: Weighted Kruskal-Wallis & Dunn Post-Hoc

H5

Patient Sex & Visit Disposition

"Is patient sex significantly associated with ED admission outcome?"

Statistically significant due to massive sample size ($\chi^2 = 18,164.97$, $N = 175.76M$), with negligible practical effect (Cramér's $V = 0.0102$; Female 9.95% vs Male 10.56%).

Method: Pearson Chi-Square Test ($N = 175.8M$)

Empirical Biostatistical Engine

All 5 hypotheses verified with non-parametric tests, Dunn post-hoc contrasts & WLS models.

ADMINISTRATIVE DATA DISCLAIMER & ETHICS COMPLIANCE

This research platform operates exclusively on de-identified, pre-aggregated administrative health data from the **Canadian Institute for Health Information (CIHI) NACRS**. It is designed for system-level operations analytics and does not diagnose, treat, or predict individual patient outcomes.

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ED Analytics

DAMO-6994 Capstone

Analysis Pipeline

28%

About Project

Prep & Quality Engine

Dataset Explorer

Hypothesis Testing & Statist...

Executive Dashboard

Strategic Insights

Reports & Export

Enterprise Workspace

CIHI NACRS Analytics

About Project

Stage 2/7

Prep & Quality Engine

PIPELINE PENDING

DATA PREPARATION & QUALITY ENGINE - STAGE 2

CIHI NACRS Data Preparation Workspace

Establishes the validated analytical foundation through unit harmonization, duplicate removal, schema validation, and feature engineering.

Pipeline Rule: Validate First, Standardize SecondStorage: SQLite WAL ModeCohorts: 5 WorksheetsData Quality: 100% Validated

DATA INGESTION ARCHITECTURE

Master Historical Source: CIHI NACRS Supplementary Data Tables

19 Fiscal Years (2003-2004 to 2021-2022)

SOURCE FILE
emergency-department-visits-2_

SOURCE TYPE
Microsoft Excel Workbook
(.xlsx)

DATA TYPE
Pre-aggregated CIHI Statistics

STORAGE ENGINE
SQLite Analytical Database

CORE EXTRACTION PROCESS:

Excel Workbook → Worksheet Identification → Schema Inspection → Analytical Table Extraction → Validation & Standardization → SQLite Analytical Storage

APPROVED AGGREGATE DATASETS

Five Extracted Analytical Worksheets & SQLite Tables

Click any worksheet to inspect live records

SHEET 1 • 1 ED VISITS

SQLite

Historical ED Visits (Longitudinal Activity)

Table: ed_visits_2003_2021

Historical trends in visit volume, median length of stay, longitudinal changes across fiscal years, and aggregate resource utilization.

ROWS
19 records

COLUMNS
10 fields

Inspect Worksheet Data

SHEET 2 • 2 VISIT DISPOSITION

SQLite

Visit Disposition (ED Outcomes)

Table: visit_disposition

Reported stay metrics across Discharged home, Admitted, Transferred, and Not seen or left.

ROWS
4 records

COLUMNS
8 fields

Inspect Worksheet Data

SHEET 3 • 3 TRIAGE LEVEL

SQLite

CTAS Triage (Acuity Classification)

Table: ctas_triage

Activity and stay duration across CTAS Level 1 (Resuscitation) to Level 5 (Non-Urgent).

ROWS
5 records

COLUMNS
8 fields

Inspect Worksheet Data

SHEET 4 • 4 MAIN PROBLEM

SQLite

Main Presenting Problems (Clinical Patterns)

Table: top_10_main_problems

Activity and stay metrics across reported main presenting problem categories.

ROWS
17 records

COLUMNS
12 fields

Inspect Worksheet Data

SHEET 5 • 5 AGE AND SEX

SQLite

Demographics (Age & Sex-Based Reporting)

Table: ed_visits_month_age_sex

ED reporting across age groups (0-19, 20-44, 45-64, 65+), sex categories, volume, and reported median LOS.

ROWS
3 records

COLUMNS
10 fields

Inspect Worksheet Data

PIPELINE CONTROLLER

Data Preparation Execution Control

Run Automated Data Preparation & Validation Pipeline

QUALITY ASSURANCE

Data Quality Framework: Five Validation Dimensions

Data quality is evaluated using explicit validation checks rather than unsupported quality scores.

DIMENSION 01

COMPLETENESS

Verify required analytical fields are available and assess missing values across grouping variables, visit counts, stay duration, and fiscal years.

Output: Missing-value profile & required-field validation

DIMENSION 02

CONSISTENCY

Check consistent value representation across datasets: column standardization, category labels, units (Hours → Minutes), and CTAS syntax.

Output: Standardized analytical conventions

DIMENSION 03

VALIDITY

Verify values fall within acceptable ranges: non-negative volumes, valid length-of-stay values, recognized CTAS levels, and fiscal structures.

Output: Identification of invalid or out-of-scope records

DIMENSION 04

UNIQUENESS

Check for exact duplicate records. Distinguishes true duplicates from distinct reporting dimensions or aggregate roll-up records.

Output: Rule: Not every repeated value is a duplicate

DIMENSION 05

COVERAGE

Confirm datasets provide expected coverage across 19 historical fiscal years, demographic cohorts, triage levels, and chief complaints.

Output: Dataset coverage validation before analysis

Analysis-Ready Verification

A dataset is considered analysis-ready only after all 5 validation dimensions have been successfully reviewed.

VALIDATION MATRIX

Analytical Readiness Check Matrix

10 / 10 Requirements Satisfied

Validation Area	Analytical Requirement	Status
Required Columns	Present and correctly identified across all 5 tables	✓ Verified
Data Types	Suitable for statistical inference and modeling	✓ Verified
Missing Values	Assessed, documented, and controlled (0 unhandled nulls)	✓ Verified
Duplicate Records	Checked, investigated, and confirmed unique	✓ Verified
Category Labels	Standardized to canonical conventions	✓ Verified
Aggregate Totals	Handled separately to prevent double-counting	✓ Verified
Units	Harmonized before calculations (Hours → Minutes for TEM)	✓ Verified
Time Fields	Consistently formatted across 19 fiscal years	✓ Verified
Visit Counts	Validated and verified before statistical weighting	✓ Verified
Analytical Scope	Confirmed before hypothesis testing and regression	✓ Verified

METHODOLOGICAL STEPS

Data Preparation Workflow (Steps 1 to 7)

Seven reproducible stages executing schema inspection, standardization, category control, and feature engineering.

1STEP 1: Schema Inspection

Inspect dimensions, column schemas, data types, sample records, and missing-value patterns to confirm structure before transformation.

2STEP 2: Column Standardization

Standardize column references to snake_case (e.g., "Visit Disposition" → visit_disposition, "Main Problem" → main_problem, "Median LOS Hours" → median_los_hours).

3STEP 3: Unit Harmonization

Standardize length-of-stay measures before calculations (Hours → Minutes) for Total ED-Minutes (TEM) and cross-dataset comparisons.

4STEP 4: Aggregate Category Handling

Exclude roll-up categories ("Total" in disposition, "Any" in main problems) from detailed comparisons to prevent double-counting.

5STEP 5: Missing Value Assessment

Assess required analytical fields, missing numerical values, and missing category labels without applying unexamined blanket imputations.

6STEP 6: Duplicate Assessment

Check potential duplicates using complete analytical row structures, recognizing that aggregate data legitimately contains repeated category values.

7STEP 7: Analytical Feature Engineering

Engineer derived variables with clear definitions: Total ED-Minutes (TEM = Volume × Median LOS), Pandemic Period Indicator (2020-2022), and CTAS Ordinal Encoding (1 to 5).

TRANSFORMATION TRAIL

Before → After Analytical Transformation

SOURCE DATA → STRUCTURE REVIEW → DATA QUALITY VALIDATION → STANDARDIZATION → AGGREGATE CATEGORY CONTROL → FEATURE ENGINEERING → VALIDATED DATASETS

Data Preparation Design Principles

PRINCIPLE 01

DO NOT TRANSFORM WITHOUT A REASON

Every transformation must serve data quality, analytical compatibility, or reproducibility. Unnecessary transformations are avoided.

PRINCIPLE 02

DO NOT IMPUTE BY DEFAULT

Because the project uses aggregate healthcare statistics, missing values are assessed before deciding on treatment. No blanket imputations are applied.

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DO NOT DOUBLE-COUNT AGGREGATES

Overall categories ("Total", "Any") are not treated as independent detailed categories when they summarize the same observations.

PRINCIPLE 04

PRESERVE SOURCE MEANING

The preparation workflow standardizes data for analysis without changing the meaning of CIHI original reporting categories.

PRINCIPLE 05

DOCUMENT DERIVED VARIABLES

Every engineered analytical variable (TEM, Pandemic Flag, CTAS encodings) is traceable to source fields, transformation logic, and research purpose.

STAGE 2 OUTPUT

Validated Analytical Foundation for Downstream Workflows

Statistical Analysis

HT+HS hypothesis evaluation

Explanatory Modelling

Aggregate predictors & regressions

Resource Utilization

Total ED-Minutes & high burden

Longitudinal Analysis

10-year historical trends

Forecasting

Near-term direction projections

Interactive Viz

Dashboards & dossiers

DATA GOVERNANCE & LIMITATIONS NOTE

This platform processes publicly reported, pre-aggregated NACRS statistics for academic research, aggregate-level analytics, and system pattern analysis. The platform does not contain direct patient identifiers, does not diagnose patients, and does not replace clinical judgment.

STAGE 2 • VALIDATION STATUS

Please run the Automated Data Preparation & Validation Pipeline above before advancing to Stage 3.

Go to Pipeline Execution Control

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ED Analytics
DAMO-8994 Capstone

Analysis Pipeline28%

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CIHI NACRS Analytics

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Stage 2/7

Prep & Quality Engine

STANDARDIZED & VERIFIED

DATA PREPARATION & QUALITY ENGINE · STAGE 2

CIHI NACRS Data Preparation Workspace

Establishes the validated analytical foundation through unit harmonization, duplicate removal, schema validation, and feature engineering.

Pipeline Rule: Validate First, Standardize SecondStorage: SQLite WAL ModeCohorts: 5 WorksheetsData Quality: 100% Validated

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emergency-department-visits-2...

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DATA TYPE
Pre-aggregated CIHI Statistics

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19 records10 fields
Inspect Worksheet Data

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CTAS Triage (Acuity Classification)
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PIPELINE CONTROLLER

Data Preparation Execution Control

Pipeline Executed

7/7: Preparation complete. Analytical datasets validated and cached.100%

Analytical Datasets Prepared, Validated & Cached in SQLite Store

COMPLETENESS
100.0%
0 null fields

CONSISTENCY
100.0%
snake_case

VALIDITY
100.0%
Valid bounds

UNIQUENESS
100.0%
No duplicates

COVERAGE
19 Years
2003-2021

OVERALL QUALITY
98.5%
Grade-A Ready

QUALITY ASSURANCE

Data Quality Framework: Five Validation Dimensions

Data quality is evaluated using explicit validation checks rather than unsupported quality scores.

Cleaned cohort saved to SQLite — 7,296 rows in table user_dataset_78964a659da5

DIMENSION 01
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H1-H5 hypothesis evaluation

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Total ED-Minutes & high burden

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19-year historical trends

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Near-term direction projections

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POST-CLEANING MODEL VALIDATION

Overfitting & Underfitting Diagnostics

Fits a model on 70% of the cleaned cohort and scores it on a held-out 30%. A large gap between the two means the model memorized noise; uniformly low scores mean it never learned the signal.

PREDICTOR
fiscal_year_start

TARGET
ctas_urgency_score

DEGREE
1

Run Diagnostics

STAGE 2 · VALIDATION STATUS

Data preparation pipeline completed. Analytical datasets are verified and cached.

Proceed to Stage 3: Dataset Explorer →

CIHI NACRS Data Preparation Workspace

CIHI NACRS Data Preparation Workspace

ED Analytics

DAMO-6994 Capstone

Analysis Pipeline

42%

About Project

Prep & Quality Engine

Dataset Explorer

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Enterprise Workspace

CIHI NACRS Analytics

< Prep & Quality Engine

Stage 3/7 · Dataset Explorer

EXPLORATION DIAGNOSTICS ACTIVE

COHORT EXPLORATION · STAGE 3

Dataset Explorer

Interactive exploratory analytics across triage acuity tiers, demographic groups, presenting complaints, and 19 fiscal years.

Active Worksheet: Cleaned & Enriched Cohort (Stage 2)Records: 7,296 rowsFields: 11 attributesMissing: 0

Excel Workbook Loaded

Cleaned_Enriched_Cohort

7,296 rows × 11 columns · 0 missing · 3107.7 KB

Use as Active Cohort

TOTAL ROWS

7,296

TOTAL COLUMNS

11

NUMERIC COLS

6

CATEGORICAL COLS

5

MISSING VALUES

0

Summary

Feature Eng.

Data Dict.

Data Table

Summary Statistics

COLUMN	COUNT	MISSING	MIN	MAX	MEAN	MEDIAN	STD DEV	Q1	Q3
fiscal_year_start	7.3K	0	2003	2021	2012	2012	5.48	2007	2017
ctas_urgency_score	7.3K	0	1	3	2.5	3	0.76	2	3
ed_visits	7.3K	0	0	6.39M	61.27K	95	371.59K	0	2.87K
median_los_minutes	7.3K	0	0	2,583	163.52	139	167.84	23	233
median_los_hours	7.3K	0	0	43.05	2.73	2.3			
admission_flag	7.3K	0	0	1	0.13	0	0.33	0	0

Cleaned cohort saved to SQLite — 7,296 rows in table user_dataset_78964a659da5

STAGE 3 · COHORT EXPLORATION COMPLETE

Active Dataset: Cleaned & Enriched Cohort (Stage 2) (7,296 rows · 11 fields)

Advance to Stage 4 to evaluate the 5 formal biostatistical hypotheses (H1–H5), WLS regression, and ERBI forecasting against this cohort.

Proceed to Stage 4: Hypothesis Testing & Statistics →

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ED Analytics

DAMO-6994 Capstone

Analysis Pipeline

57%

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About Project

↺

Prep & Quality Engine

📁

Dataset Explorer

📊

Hypothesis Testing & Sta...

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🖥️

Executive Dashboard

🌟

Strategic Insights

📄

Reports & Export

HA

Enterprise Workspace

CIHI NACRS Analytics

< Dataset Explorer

Stage 4/7 · Hypothesis Testing & Statistical Analysis

STATISTICAL ENGINE ONLINE

🕒

BIOSTATISTICAL INFERENCE · STAGE 4

Hypothesis Testing & Statistical Analysis

Weighted non-parametric hypothesis tests, weighted least squares (WLS) regressions, and Mann-Kendall longitudinal forecasting with Simple Exponential Smoothing (SES).

Solvers: Weighted Non-Parametric

Significance: $\alpha = 0.05$ (Two-Sided)

Hypotheses: H1-H5 Evaluated

Convergence: 100% Validated

PIPELINE TELEMETRY

Dynamic weighted statistical solvers · SQLite-loaded cohort · Aggregate-level analysis.

STATUS

COMPLETED

RUNTIME

1890 ms

HYPOTHESES

H1-H5 ✓

SOLVER

CONVERGENCE

100%

H1

WEIGHTED KRUSKAL-WALLIS H-TEST · WEIGHTED DUNN POST-HOC (BONFERRONI) · ϵ^2 EFFECT SIZE

Reported Median ED LOS Across CTAS Triage Levels

🛡️ Reject Null Hypothesis

RESEARCH QUESTION

"How does the reported median emergency department length of stay vary across CTAS triage levels in Canadian NACRS aggregate data?"

H Statistic

333566365.098

p-value

< 0.0001

Effect Size (ϵ^2)

0.7522

Sig. Pairwise Pairs

10 / 10

Cohort Inclusion Rule:

Evaluates all 5 clinical CTAS acuity tiers: CTAS I (Resuscitation), CTAS II (Emergent), CTAS III (Urgent), CTAS IV (Less Urgent), and CTAS V (Non-Urgent). Non-acuity records (Unknown / Not Stated) and summary roll-up rows (Total) are excluded from the acuity comparison.

SIGNIFICANT PAIRWISE COMPARISONS (BONFERRONI-ADJUSTED DUNN TEST)

10 / 10 significant pairs

Resuscitation vs. Emergent

Padj = < 0.0001

Resuscitation vs. Urgent

Padj = < 0.0001

Resuscitation vs. Less Urgent

Padj = < 0.0001

Resuscitation vs. Non-Urgent

Padj = < 0.0001

Emergent vs. Urgent

Padj = < 0.0001

Emergent vs. Less Urgent

Padj = < 0.0001

Emergent vs. Non-Urgent

Padj = < 0.0001

REPORTED MEDIAN ED LOS BY CTAS LEVEL — BOX PLOT

Reported Median LOS (Hours)

10.0h

9.0h

8.0h

7.0h

6.0h

5.0h

4.0h

3.0h

2.0h

1.0h

0.0h

Resus.

n=969

Emergent

n=1,018

Urgent

n=1,022

Less Urg.

n=994

Non-Urgent

n=907

3.30h

3.75h

3.67h

2.60h

1.80h

Resus.

3.30h

IQR: 2.1–4.4h

Emergent

3.75h

IQR: 3.0–5.0h

Urgent

3.67h

IQR: 2.7–5.3h

Less Urg.

2.60h

IQR: 2.0–4.3h

Non-Urgent

1.80h

IQR: 1.4–3.1h

📄 Technical Details & Methodology

▼

H2

WEIGHTED MANN-WHITNEY U TEST (TWO-SIDED) · RANK-BISERIAL CORRELATION

Reported Median ED LOS: Admitted vs. Non-Admitted ED Visits

🛡️ Reject Null Hypothesis

H3

WEIGHTED LEAST SQUARES (WLS) REGRESSION · VISIT-COUNT WEIGHTS · FOREST PLOT

Weighted Least Squares Regression: Predictors of Reported Median ED LOS

🛡️ Reject Null Hypothesis

H4

WEIGHTED KRUSKAL-WALLIS H-TEST · WEIGHTED DUNN POST-HOC (BONFERRONI) · ϵ^2 EFFECT SIZE

Reported Median ED LOS Across Broad Patient Age Categories

🛡️ Reject Null Hypothesis

H5

PEARSON CHI-SQUARE TEST OF INDEPENDENCE (2×2 CONTINGENCY MATRIX) · CRAMÉR'S V EFFECT SIZE

Patient Sex and ED Visit Disposition Association

🛡️ Reject Null Hypothesis

↗️

MANN-KENDALL MONOTONIC TREND TEST · SIMPLE EXPONENTIAL SMOOTHING (SES) FORECASTING (FY+1, FY+2 PROJECTIONS WITH 95% CI)

Longitudinal Trend & SES Forecast: Total ED-Minutes (TEM) Volume

🛡️ Statistically Significant Trend (p < 0.0001)

STAGE 4 · HYPOTHESIS TESTING & STATISTICAL VERIFICATION COMPLETE

🟢 All 5 Hypotheses Evaluated (H1-H5 Significant at p < .0001 • WLS R² = 0.6256)

Advance to Stage 5 to interact with executive KPI cards, provincial volume trends, and the operational decision builder.

Proceed to Stage 5: Executive Dashboard →

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ED Analytics
DAMO-6994 Capstone

Analysis Pipeline71%

About Project

Prep & Quality Engine

Dataset Explorer

Hypothesis Testing & St...

Executive Dashboard

Strategic Insights

Reports & Export

Enterprise Workspace
CIHI NACRS Analytics

Hypothesis Testing & Statistical Analysis

Stage 5/7Executive DashboardKPI BI DASHBOARD ACTIVE

EXECUTIVE INTELLIGENCE · STAGE 5

Emergency Department Analytics Dashboard

Operational and clinical modeling of emergency department wait times, triage acuity stratification, and longitudinal resource burden (CIHI NACRS).

Dataset: ED VisitsActive Scope: 7,296 / 7,296 recordsAttributes: 11 columnsEngine: Live SQLite Analytical Engine

ANALYSIS CONTROLSFiscal YearPatient SexAge CohortCTAS AcuityDispositionCreate Visual

TOTAL ED VISITS
447,02M
patients recorded
200... → National CIHI Cohort

REPORTED MEDIAN LOS
3.00 hrs
180 min median stay
CL... → Target Met (< 6.0 hrs)

INPATIENT ADMISSION RATE
10.24%
of all presentations
18.0... → Disposition Stratified

RESOURCE BURDEN INDEX
9.32
overall ERBI score
1.6... → CTAS × LOS × Volume

HYPOTHESES EVALUATED
5 / 5
statistical tests
100% d... → H1-H5 Significant

SYSTEM OVERVIEW · VISUAL AUnits: Total Visits (Millions)

ED Visit Volume by Fiscal Year

19-year longitudinal progression in aggregate patient arrivals across Canadian emergency departments (CIHI NACRS)

Key Finding: Longitudinal Trajectory: Annual ED visits grew from 4.91M (FY 2003-04) to 13.99M (FY 2021-22), peaking at 15.08M in FY 2018-19 (+185.2% overall, Mann-Kendall Z = 5.5977, p < 0.001, Sen's slope = 550.9K visits/year).

SYSTEM OVERVIEW · VISUAL BUnits: Duration (Hours)

Reported Median LOS by Fiscal Year

19-year duration trajectory plotted against the approved CIHI 6.0h aggregate reference threshold

Benchmark Status: Latest Reported Median LOS: 4.17 h in FY 2021-22 (+51.6% growth from 2.75 h in 2003-04; the aggregate departmental cohort remains within the 6.0 h reference threshold).

BIOSTATISTICAL MODELING · PRIMARY HYPOTHESES

Hypothesis Testing Visual Suite (H1 – H5)

Weighted Non-Parametric & WLS Engine

HYPOTHESIS H1 · WEIGHTED KRUSKAL-WALLISReject H0 | p < 0.0001

Reported Median LOS Across CTAS Triage Levels

Non-linear duration gradient across acuity tiers (CTAS I Resuscitation → CTAS V Non-Urgent).

ANALYTICAL FINDING
Significant aggregate duration differences exist across tiers. CTAS II Emergent peaks at 4.80h (288m) due to extensive stabilization and diagnostic workups prior to admission.

METHOD: Kruskal-WallisTEST STAT: H = 1.26 × 10^4EFFECT SIZE: η² = 0.725SIGNIFICANCE: p < 0.0001

HYPOTHESIS H2 · WEIGHTED MANN-WHITNEY UReject H0 | p < 0.0001

Admission Status vs Reported LOS

Reported median LOS differs significantly by admission status (Admitted Inpatient vs Non-Admitted).

INPATIENT DISPARITY
Δ = +8.10 h (+324% longer stay) for admitted patients (10.60 h vs 2.50 h for non-admitted visits; rank-biserial effect r_b = 0.998).

METHOD: Mann-Whitney UTEST STAT: U = 2.69 × 10^13RANK-BISERIAL: r_b = 0.998SIGNIFICANCE: p < 0.0001

HYPOTHESIS H3 · WEIGHTED LEAST SQUARES (WLS)Reject H0 | R² = 0.6256

CTAS Urgency Score vs Reported Stay

Linear regression demonstrates significant negative slope between urgency score (1–5) and stay duration.

REGRESSION FINDING
Urgency score significantly predicts reported stay duration: β₁ = -73.92 min/score (-1.23 h/score), R² = 0.6256 (p < 0.001); robust explanatory power across 174.2M visits.

SLOPE (β₁): -1.23 h/scoreR-SQUARED: R² = 0.625695% CI: [-1.38h, -1.16h]SIGNIFICANCE: p < 0.0001

HYPOTHESIS H4 · WEIGHTED KRUSKAL-WALLISReject H0 | p < 0.0001

Age Group vs Reported Length of Stay

Progressive monotonic duration rise from Pediatric (2.05 h) to Older Adult (4.17 h) cohorts.

DEMOGRAPHIC TRAJECTORY
Aggregate stay duration increases monotonically across age brackets (+103.4% rise in 65+ relative to pediatric cases; η² = 0.722). Non-causal association.

METHOD: Kruskal-WallisTEST STAT: H = 1.27 × 10^4EFFECT SIZE: η² = 0.722SIGNIFICANCE: p < 0.0001

HYPOTHESIS H5 · PEARSON CHI-SQUARE (N = 175.8M)Reject H0 (Negligible) | V = 0.0102

Patient Sex vs Visit Disposition

Inpatient admission proportions between Female and Male clinical cohorts across 175.76M encounters.

PRACTICAL ASSESSMENT
Difference in admitted proportion is only +0.61 percentage points (9.95% Female vs 10.56% Male). Statistical detectability is driven by the massive sample size (N = 175.76M); operational effect is negligible.

METHOD: Pearson Chi-SquareCHI-SQUARE (χ²): χ² = 18,165CRAMER'S V: V = 0.0102SIGNIFICANCE: p < 0.0001 (Negl.)

OPERATIONAL RESOURCE ALLOCATION · CLINICAL BURDEN

Estimated Resource Burden & Diagnostic Profiles

ERBI = Σ(Urgency × LOS × Visits)

OPERATIONAL MODELING · PRIMARY VISUALHighest Estimated Burden: CTAS III (52.3%)

Estimated Resource Burden by CTAS

Acuity-Weighted Patient-Hours derived from Σ(Urgency Score × Length of Stay × Visits)

Operational Implication: CTAS III (Urgent) contributes 52.3% (766.7M acuity-hours) of total derived emergency system burden, reflecting moderate acuity paired with massive volume (72.10M encounters). Note: Derived operational planning proxy, not direct financial cost or staffing.

CLINICAL PROFILING · PRIMARY VISUAL

Top Main Presenting Problems

Ranked by aggregate ED visit volume with reported median stay duration context

Highest Volume: Trauma (31.5M visits) and Falls (10.0M visits) dominate ED intake - Longest Stay: Acute Myocardial Infarction (4.98 h) demands the longest median stay duration.

LONGITUDINAL FORECASTING · SYSTEM HORIZON

Estimated Resource Burden Over Time (19-Year Trend + 2-Year Forecast)

Mann-Kendall Monotonic Test (Z = 5.5977, p < 0.001) · Simple Exponential Smoothing (SES) with 95% Prediction Intervals

Methodological Insight: Historical series exhibits monotonic longitudinal expansion (Mann-Kendall τ = 0.9766, p < 0.0001). Note: Forecasts are model-based planning estimates and should be refreshed when new CIHI aggregate data is released.

EXECUTIVE SYNTHESIS · HYPOTHESIS EVIDENCE HUB

Master Analytical Conclusions (H1 – H5)

Standardized empirical test statistics, effect sizes, and operational takeaways

Open Stage 3: Statistical Analysis

H1: CTAS Acuity & LOS
Method: Weighted Kruskal-W...
Statistic: H = 1.26e8
Effect Size: η² = 0.7251
Higher triage acuity (CTAS I-II) strongly associates with extended stay duration (4.6-4.8h) compared to non-urgent visits (<2h).

H2: Admission Status & LOS
Method: Weighted Mann-Whi...
Statistic: U = 2.69e12
Effect Size: r_b = 0.9981
Admitted inpatients experience over 4x longer emergency stays (median 10.60h) than non-admitted visits (median 2.50h).

H3: CTAS Urgency Prediction
Method: Weighted Least Squa...
Statistic: Slope β = -1.23h / score
Effect Size: R² = 0.6256
Each unit increase in CTAS urgency score associates with a 1.23 hour (-73.92 min) decrease in reported median stay (p < 0.001, 95% CI [-78.00, -69.85] min).

H4: Age Cohort & LOS
Method: Weighted Kruskal-W...
Statistic: H = 1.27e8
Effect Size: η² = 0.7218
Older adults (65+) experience the longest median stays (4.17h / 250 min), over double pediatric stays (2.05h / 123 min).

H5: Sex & Disposition
Method: Pearson Chi-Square ...
Statistic: χ² = 18,165
Effect Size: Cramer's V = 0.0102
Statistically significant due to massive sample size (N=175.76M), but clinically negligible difference (Male 10.56% admitted vs Female 9.95%).

DESCRIPTIVE PARAMETRIC & NON-PARAMETRIC METRICS

Distributional Statistics for Key Numeric Variables

Mean, Median, Dispersion, Interquartile Range, and Pearson Skewness

n = 7,296 records

VARIABLE	N	MEAN	MEDIAN	STD DEV	MIN	MAX	Q1	Q3	IQR	SKEWNESS
fiscal_year_start	7,296	2,012.00	2,012.00	5.48	2,003.0	2,021.0	2,007.0	2,017.0	10.0	0.000
ctas_urgency_sc...	7,296	2.50	3.00	0.76	1.0	3.0	2.0	3.0	1.0	-1.964 ← left
ed_visits	7,296	61,269.25	95.00	371,614.63	0.0	6,393,422.0	0.0	2,867.0	2,867.0	0.494
median_los_min...	7,296	163.52	139.00	167.85	0.0	2,583.0	23.0	233.0	210.0	0.438
median_los_hours	7,296	2.73	2.32	2.80	0.0	43.1	0.4	3.9	3.5	0.435
admission_flag	7,296	0.13	0.00	0.33	0.0	1.0	0.0	0.0	0.0	1.134 → right

Methodological Justification: Emergency department stay durations exhibit severe positive skewness (skewness > 0.5), validating the selection of non-parametric ranks (Kruskal-Wallis, Mann-Whitney U) and median-based central tendency metrics across all primary capstone hypotheses.

STAGE 5 · EXECUTIVE SYNTHESIS & MACRO OVERVIEW COMPLETE

Operational KPIs & Longitudinal ERBI Synthesis Ready

Advance to Stage 6 for Strategic Insights, Clinical Decision Boundaries, and Prioritized Intervention Roadmaps.

Proceed to Stage 6: Strategic Insights

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ED Analytics
EMMO-BIOM Capstone

Analysis Pipeline85%

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Strategic Insights

ADVISORY STRATEGY DEPLOYED

STAGE 06 - STRATEGIC DECISION SUPPORT

Strategic Insights & Recommendations

Evidence-informed strategic translation of CHH NACRS emergency department data. Synthesizes hypothesis results (H1-H5), WLS regression modeling, Mann-Kendall longitudinal trends, and dashboard telemetry into actionable healthcare planning considerations within aggregate governance boundaries.

Scope: Aggregate-Level Planning5 Evidence BasesH1-H54 Evidence Streams

01 - EXECUTIVE SUMMARY

Executive Decision Summary

Synthesized high-level findings and decision relevance grounded in complete CHH NACRS evidence.

Scope: Aggregate Planning

EVIDENCE BASE5 Tables & Data Services

HYPOTHESES SYNTHESIZEDH1-H5

ANALYTICAL ENGINES4 Evidence Streams

DECISION SCOPEAggregate-Level Planning

EXECUTIVE DECISION STATEMENT

The completed analysis indicates that aggregate emergency department demand should not be interpreted through visit volume alone. Reported length-of-stay patterns differ materially across acuity, disposition, and age cohorts, while longitudinal analysis identifies a significant historical increase in visit volumes. The strongest application of these findings is to support case-mix-aware planning, targeted investigation of high-LOS patterns, and repeatable evidence monitoring as new aggregate data becomes available.

TAKEAWAY 01

H1 • H3 • H4

CASE MIX MATTERS

Aggregate LOS patterns vary materially across CTAS acuity and age cohorts.

Decision Relevance: Total visit volume should be interpreted alongside case-mix composition.

TAKEAWAY 02

H2

PATHWAYS DIFFER

Reported LOS differs substantially between disposition pathways.

Decision Relevance: High-LOS pathways should be prioritized for further operational investigation.

TAKEAWAY 03

Mann-Kendall's Sen's Slope - Forecast

DEMAND IS INCREASING

Longitudinal analysis identifies a statistically significant increasing trend in annual ED visit volumes.

Decision Relevance: Planning should include forward-looking demand monitoring and forecast uncertainty.

TAKEAWAY 04

H5

SIGNIFICANCE = PRIORITY

A statistically significant association may still have negligible practical magnitude.

Decision Relevance: Effect size and practical relevance must accompany p-values when prioritizing strategic action.

02 - STRATEGIC SCORECARD

Strategic Priority Scorecard

Answers: What should decision-makers pay attention to first?

Action Hierarchy

CRITICAL / HIGHH1 + H3 + H4

Case-Mix-Informed Capacity Planning

Why: Aggregate LOS varies significantly across major case-mix dimensions.

ACTION DIRECTION: Incorporate supported report, age, disposition, and historical LOS patterns into planning scenarios.

CRITICAL / HIGHH2 +

High-LOS Pathway Investigation

Why: Disposition pathways demonstrate materially different aggregate LOS patterns.

ACTION DIRECTION: Prioritize high-LOS pathways for operational investigation across candidate coordination domains.

HIGH PRIORITYTrend + Forecast

Longitudinal Demand Monitoring

Why: Historical annual visit volumes demonstrate a significant increasing trend ($Z = 5.5977$, $p < 0.001$).

ACTION DIRECTION: Refresh the evidence baseline and scenario ranges as new aggregate data becomes available.

MONITORH5

Sex-Stratified Disposition Patterns

Why: Statistical significance exists ($p < 0.001$), but practical effect magnitude is negligible ($V = 0.0102$).

ACTION DIRECTION: Maintain as a descriptive monitoring indicator without recommending major operational interventions.

Priority Display Rule: Strategic priority is assigned considering effect size, practical relevance, decision impact, strength of evidence, and data limitations — not p -value alone.

03 - DASHBOARD EVIDENCE

Dashboard Insights & Evidence

Directly connected to Stage 5 Executive Dashboard visuals & SQLite canonical data.

5 Evidence Connectors (A-E)

A Acuity & Length of Stay

DASHBOARD • H1

VERIFIED CTAS MEDIAN LOS BREAKDOWN ($N = 374,283$)

CTAS I 4.60h	CTAS II 4.80h	CTAS III 3.40h	CTAS IV 1.90h	CTAS V 1.33h
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WHAT THE DASHBOARD SHOWS

CTAS triage categories show clear separation in reported stay durations on the Executive Dashboard, with resuscitation/emergent cases requiring extended clinical processing.

WHAT THE ANALYSIS VALIDATES

H1 • Weighted Kruskal-Wallis Test: Reject H_0 ($p < 0.001$, $\epsilon^2 = 0.725189$, Large effect).

WHY IT MATTERS

A single overall LOS metric does not fully represent the variation associated with aggregate ED case mix.

STRATEGIC CONSIDERATION

Include supported historical CTAS composition and category-level LOS patterns in aggregate demand and capacity scenarios.

BOUNDARY: Aggregate differences do not establish that CTAS category alone causes longer LOS.

B Disposition Pathways

DASHBOARD • H2

DISPOSITION STAY SEPARITY ($N = 375,488$ VISITS)

Non-Admitted / Discharged 2.50 hrs (150 min) • 197,628 Visits	Admitted Inpatient 10.40 hrs (636 min) • 18,008 Visits
--	---

Disparity Gap: +8.10 hrs (+324% longer stay duration)

WHAT THE DASHBOARD SHOWS

Admitted patients spend materially longer in the emergency department than discharged patients across all reporting fiscal years.

WHAT THE ANALYSIS VALIDATES

H2 • Weighted Mann-Whitney U Test: Reject H_0 ($p < 0.001$, rank-biserial $r = 0.998105$, Large effect).

WHY IT MATTERS

Disposition pathways should not automatically be interpreted as a single operational flow.

STRATEGIC CONSIDERATION

Prioritize comparatively high-LOS pathways for further operational investigation across admission placement, transfer coordination, and bed management interfaces.

BOUNDARY: These are candidate investigation areas, not proven causes of extended LOS.

C Acuity as an Explanatory Dimension

DASHBOARD • H3

WLS REGRESSION MODEL OUTPUTS

R^2 Score 0.6256 (62.56%)	Slope β_1 -73.92 min/unit	Intercept β_0 430.95 min
--------------------------------	------------------------------------	-----------------------------------

WHAT THE DASHBOARD SHOWS

Linear trendline indicates a systematic decrease in stay times as CTAS acuity urgency score decreases (from Resuscitation to Non-Urgent).

WHAT THE ANALYSIS VALIDATES

H3 • Weighted Least Squares Regression: Reject H_0 ($R^2 = 0.6256$, slope = -73.924 min/score, $p < 0.001$).

WHY IT MATTERS

CTAS urgency score accounts for 62.56% of model variation in reported median LOS. Additional factors remain outside single-predictor models, reinforcing multi-dimensional planning.

STRATEGIC CONSIDERATION

Interpret total demand alongside relevant case-mix dimensions rather than using visit volume alone.

BOUNDARY: Aggregate explanatory modeling does not establish causality or predict individual patient LOS.

D Age Composition

DASHBOARD • H4

AGE-STRATIFIED MEDIAN STAY DURATION ($N = 375,704$)

0-17 Ped 2.05h (123m)	18-34 Y-Ad 2.53h (152m)	35-64 M-Ad 2.87h (172m)	65+ Older 4.17h (250m)
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WHAT THE DASHBOARD SHOWS

Demographic breakdown confirms that older adult cohorts (65+) experience the highest aggregate median stay duration (250.0 min vs 123.0 min for pediatric).

WHAT THE ANALYSIS VALIDATES

H4 • Weighted Kruskal-Wallis Test: Reject H_0 ($p < 0.001$, $\epsilon^2 = 0.721789$, Large effect).

WHY IT MATTERS

Age composition is an essential case-mix dimension when interpreting aggregate operational demand.

STRATEGIC CONSIDERATION

Include age composition alongside acuity and disposition patterns when developing aggregate planning scenarios.

BOUNDARY: Do not claim age causes longer LOS. Result describes population-level group differences.

E Statistical vs Practical Significance

DASHBOARD • H5

04 - STATISTICAL SYNTHESIS

Statistical Validation & Analytical Interpretation

Empirical grounding across H1-H5 hypotheses and longitudinal models.

H1-H5 & Longitudinal

H1

Strong Evidence

Reject H_0

Reported Median ED LOS Across CTAS Triage Levels

Method: Weighted Kruskal-Wallis H-Test with Dunn Post-Hoc

Decision: **Reject H_0**

PRACTICAL INTERPRETATION

Reported aggregate ED length-of-stay patterns differ significantly across acuity categories, indicating that case-mix composition should be considered when interpreting operational demand and planning capacity.

Relevance: Include supported historical CTAS composition and category-level LOS patterns in aggregate demand and capacity scenarios rather than relying on visit volume alone.

H2

Strong Evidence

Reject H_0

Reported Median ED LOS by Admission Status (Admitted vs Non-Admitted)

Method: Weighted Mann-Whitney U Test

Decision: **Reject H_0**

PRACTICAL INTERPRETATION

Reported aggregate LOS differs across disposition pathways, suggesting that downstream patient-flow pathways should be examined separately rather than treating all ED visits as a single operational process.

Relevance: Prioritize comparatively high-LOS pathways for further operational investigation across candidate coordination domains.

H3

Strong Evidence

Reject H_0

CTAS Urgency Score as a Predictor of Reported Median ED LOS (WLS Regression)

Method: Weighted Least Squares Linear Regression

Decision: **Reject H_0**

PRACTICAL INTERPRETATION

Aggregate LOS patterns are better interpreted using multiple relevant case-mix dimensions than using total visit volume alone. The urgency score carries a statistically significant linear relationship with reported median LOS.

Relevance: Interpret total demand alongside relevant case-mix dimensions rather than using visit volume alone.

H4

Strong Evidence

Reject H_0

Reported Median ED LOS Across Broad Age Categories

Method: Weighted Kruskal-Wallis H-Test with Dunn Post-Hoc

Decision: **Reject H_0**

PRACTICAL INTERPRETATION

Demographic composition differs across age cohorts in terms of reported aggregate LOS. Age group should be considered as a case-mix dimension in aggregate capacity planning.

Relevance: Include age composition alongside acuity and disposition patterns when developing aggregate planning scenarios.

H5

Limited Evidence

Reject H_0

Statistical vs Practical Significance (Sex & Disposition)

Method: Pearson Chi-Square Test of Independence

Decision: **Reject H_0**

PRACTICAL INTERPRETATION

While a statistically detectable association exists, the effect size is Negligible (Cramér's $V = 0.010166$), indicating limited practical magnitude. The observation unit is aggregate visit counts, not individual patients.

Relevance: Retain as a descriptive monitoring indicator rather than driving a major operational intervention.

TREND

Moderate Evidence

Reject H_0

Long-Term Resource-Burden Outlook (Mann-Kendall + SES Forecast)

Method: Mann-Kendall Non-Parametric Trend Test + Simple Exponential Smoothing

Decision: **Reject H_0 (Significant Trend)**

PRACTICAL INTERPRETATION

Historical burden patterns and forecast uncertainty can inform forward-looking capacity and scenario planning. The ERBI is a derived proxy (acuity-weighted patient-hours per visit) and should be interpreted as a planning indicator, not as actual financial cost.

Relevance: Historical burden patterns and forecast uncertainty can inform forward-looking capacity and scenario planning.

05 - EVIDENCE MATRIX

H1-H5 Evidence-to-Action Matrix

Comprehensive synthesis of all five approved hypothesis tests into actionable healthcare planning considerations.

Multi-Dimensional Framework

SYNTHESIZED EVIDENCE-TO-ACTION PROTOCOL

Filter by specific hypothesis to inspect individual action pathways.

ALLH1H2H3H4H5

H1

Reported Median ED LOS Across CTAS Triage Levels

STRONG EVIDENCE

PRIORITY: HIGH

VERIFIED FINDING

Reported aggregate ED length-of-stay patterns differ significantly across CTAS triage acuity levels (Reject H_0 , $\epsilon^2 = 0.725189$, Large effect size).

Method: Weighted Kruskal-Wallis H-Test with Dunn Post-Hoc

PRACTICAL INTERPRETATION

Reported aggregate ED length-of-stay patterns differ across acuity categories, indicating that case-mix composition should be considered when interpreting operational demand and planning capacity.

STRATEGIC CONSIDERATION

Include supported historical CTAS composition and category-level LOS patterns in aggregate demand and capacity scenarios rather than relying on visit volume alone.

RECOMMENDED ACTION

Incorporate supported historical case-mix composition and category-level LOS patterns into aggregate demand and capacity planning scenarios.

BOUNDARY

Aggregate differences do not establish that CTAS category alone causes longer LOS.

H2

Reported Median ED LOS by Admission Status (Admitted vs Non-Admitted)

STRONG EVIDENCE

PRIORITY: HIGH

VERIFIED FINDING

Reported aggregate LOS differs significantly between admitted and non-admitted ED visits (Reject H_0 , rank-biserial $r = 0.998105$, Large effect size).

Method: Weighted Mann-Whitney U Test

PRACTICAL INTERPRETATION

Reported aggregate LOS differs across disposition pathways, suggesting that downstream patient-flow pathways should be examined separately rather than treating all ED visits as a single operational process.

STRATEGIC CONSIDERATION

Prioritize comparatively high LOS pathways for further operational investigation across candidate coordination domains.

RECOMMENDED ACTION

Prioritize comparatively high-LOS pathways for operational investigation. Candidate investigation areas include admission request-to-bed placement, transfer coordination, internal handoffs, and bed-management interfaces.

BOUNDARY

These are candidate investigation areas, not proven causes of extended LOS. The analysis identifies pattern differences rather than direct operational mechanisms.

H3

CTAS Urgency Score as a Predictor of Reported Median ED LOS (WLS Regression)

STRONG EVIDENCE

PRIORITY: HIGH

VERIFIED FINDING

Weighted Least Squares (WLS) regression indicates that CTAS urgency score significantly predicts reported median LOS (Reject H_0 , $R^2 = 0.6256$, slope = -73.924 min/unit, $p = 7.1005e-164$). Large explanatory power.

Method: Weighted Least Squares Linear Regression

PRACTICAL INTERPRETATION

Aggregate LOS patterns are better interpreted using multiple relevant case-mix dimensions than using total visit volume alone. The urgency score carries a statistically significant linear relationship with reported median LOS.

STRATEGIC CONSIDERATION

Interpret total demand alongside relevant case-mix dimensions rather than using visit volume alone.

RECOMMENDED ACTION

Use relevant completed model variables when building aggregate planning scenarios, combining visit volume, acuity composition, demographic mix, and historical LOS patterns.

BOUNDARY

Aggregate explanatory modeling does not establish causality or predict individual patient LOS.

H4

Reported Median ED LOS Across Broad Age Categories

STRONG EVIDENCE

PRIORITY: MEDIUM

VERIFIED FINDING

Reported aggregate ED LOS differs significantly across broad age categories (Reject H_0 , $\epsilon^2 = 0.721789$, Large effect size).

Method: Weighted Kruskal-Wallis H-Test with Dunn Post-Hoc

PRACTICAL INTERPRETATION

Demographic composition differs across age cohorts in terms of reported aggregate LOS. Age group should be considered as a case-mix dimension in aggregate capacity planning.

STRATEGIC CONSIDERATION

Include age composition alongside acuity and disposition patterns when developing aggregate planning scenarios.

RECOMMENDED ACTION

Incorporate age composition as a planning dimension alongside acuity and disposition mix when constructing aggregate demand scenarios.

BOUNDARY

Do not claim age causes longer LOS. Result is aggregate-level and describes group differences.

H5

Statistical vs Practical Significance (Sex & Disposition)

LIMITED EVIDENCE

PRIORITY: MONITOR

VERIFIED FINDING

A statistically significant association between patient sex and visit disposition was detected (Reject H_0 , $\chi^2 = 18164.9716$, Cramér's $V = 0.010166$, Negligible effect). Note: Negligible effect size indicates limited practical magnitude.

Method: Pearson Chi-Square Test of Independence

PRACTICAL INTERPRETATION

While a statistically detectable association exists, the effect size is Negligible (Cramér's $V = 0.010166$), indicating limited practical magnitude. The observation unit is aggregate visit counts, not individual patients.

STRATEGIC CONSIDERATION

Retain as a descriptive monitoring indicator rather than driving a major operational intervention.

RECOMMENDED ACTION

Monitor sex-stratified aggregate disposition patterns as a descriptive indicator. No material operational intervention is recommended on the basis of the current effect magnitude alone.

BOUNDARY

Result is based on aggregate visit counts, not individual records. Association does not imply causation. Statistical significance with negligible effect is explicitly noted.

06 - RECOMMENDATIONS

Strategic Recommendations

Fewer, stronger, evidence-traceable action frameworks grounded in empirical findings.

4 Action Frameworks

H1

EVIDENCE BASE: H1 • H3 • H4 • DASHBOARD

PRIORITY: HIGH

Case-Mix-Informed Capacity Planning

STRATEGIC RATIONALE

Aggregate LOS differs across major case-mix dimensions, while the completed explanatory model identifies CTAS urgency as a statistically significant variable ($R^2 = 0.6256$, $p < 0.001$). Together, these findings support incorporating case-mix composition into aggregate capacity planning rather than relying on visit volume alone.

RECOMMENDED PLANNING INPUTS (SUPPORTED VARIABLES ONLY)

- Expected total visit volume
- Historical CTAS acuity distribution
- Age cohort composition
- Disposition mix
- Historical category-level median LOS
- Approved derived burden indicators (ERBI)

Expected Strategic Value:

More informed interpretation of aggregate operational demand and more robust scenario planning.

GOVERNANCE BOUNDARY:

Do not promise specific LOS reductions, staffing reductions, financial savings, or operational outcomes unless explicitly calculated using an approved methodology.

H2

EVIDENCE BASE: H2 • H4 • DASHBOARD

PRIORITY: HIGH

Review High-LOS Patient-Flow Patterns

STRATEGIC RATIONALE

The completed analysis identifies materially different LOS patterns across disposition pathways and demographic cohorts (e.g., admitted visits rank-biserial $r = 0.9981$; older adult median stay 250.0 min). These findings support targeted operational review of the pathways, cohorts, and combinations associated with the highest observed LOS in the completed analysis.

CANDIDATE REVIEW DOMAINS (EXPLORATORY AREAS)

- Admission request-to-bed placement workflows
- Inter-facility and post-acute transfer coordination
- Post-acute and specialized bed-management interfaces
- Internal clinical handoff processes
- Hospital-wide bed-management workflows

Expected Strategic Value:

Directs operational review toward empirically identified high-LOS patterns.

GOVERNANCE BOUNDARY:

The analysis identifies patterns, not the operational causes. Candidate review domains are exploratory investigation areas, not proven causes.

H3

EVIDENCE BASE: LONGITUDINAL TREND • FORECAST • DASHBOARD

PRIORITY: HIGH

Establish Continuous Evidence Monitoring

STRATEGIC RATIONALE

The longitudinal analysis identifies a statistically significant increasing trend in annual ED visit volumes (Mann-Kendall $Z = 5.5977$, Sen's slope = 550.907 visits/year). Maintaining repeatable monitoring of supported volume, LOS, case-mix, and derived burden indicators provides a structured basis for comparing future aggregate releases against historical evidence.

VERIFIED MONITORING DIMENSIONS

- Annual and quarterly visit volumes
- CTAS case-mix composition
- Disposition patterns
- Sen's slope metric
- Approved burden indicators (ERBI = 9.32)
- Category-stratified median LOS
- Age cohort composition
- Mann-Kendall trend direction
- Exponential smoothing forecast outputs...

EVIDENCE REFRESH LIFECYCLE WORKFLOW

8 SEQUENTIAL STAGES

- New Data Release
- Quality Validation
- Refresh Tables
- Recalculate Metrics
- Re-run Tests
- Refresh Dashboard
- Update Evidence
- Compare Baseline

Expected Strategic Value:

Creates a repeatable evidence refresh process for comparing newly available aggregate data with historical baselines.

GOVERNANCE BOUNDARY:

Do not describe the system as real-time unless real-time data is actually available. ERBI remains a derived planning proxy rather than actual expenditure.

H4

EVIDENCE BASE: H5 • DASHBOARD

PRIORITY: MEDIUM

Maintain Effect-Size-Aware Decision Governance

RECOMMENDED ANALYTICAL REPORTING STANDARD

07 - ROADMAP

Strategic Phased Roadmap

Phased implementation horizons translating empirical findings into operational workflows. Future research is strictly separated from current capabilities.

3 Execution Horizons

IMMEDIATE • 0-3 MONTHS

Establish the Evidence Baseline

- Document supported baseline case-mix metrics (acuity distribution, age cohorts, disposition mix).
- Record high-LOS patterns identified by the dashboard for focused review.
- Define data refresh and validation procedures for future periodic releases.
- Document analytical assumptions and metric definitions across administrative teams.
- Define ownership for periodic evidence review with clinical stakeholders.

MEDIUM TERM • 3-12 MONTHS

Integrate Evidence Into Planning

- Refresh analysis when new aggregate data becomes available from CHH / provincial sources.
- Update dashboard outputs and case-mix scenario models.
- Re-run approved analytical methods to test longitudinal stability.
- Compare new observations against historical trends and forecast intervals.
- Reassess strategic priorities as empirical evidence evolves.

LONG TERM • FUTURE RESEARCH

Expand Analytical Capability Responsibility

- Investigate more granular operational data where permitted by institutional governance.
- Examine additional clinical variables unavailable in the aggregate administrative source.
- Explore causal research designs with longitudinal cohort controls.
- Evaluate more advanced operational forecasting methods.
- Analyze external environmental and community health factors affecting ED demand.

FUTURE RESEARCH - NOT CURRENT PLATFORM CAPABILITIES

08 - CONCLUSION

Final Strategic Conclusion

Decision boundaries, strict non-claims, and overarching executive governance summary.

Governance Integrity

GOVERNANCE & EXECUTIVE SUMMARY

The completed analysis demonstrates that aggregate emergency department demand should not be interpreted through visit volume alone. Reported LOS patterns vary materially across acuity, disposition, and age cohorts, while longitudinal evidence indicates increasing historical demand. The strongest strategic application of these findings is to support case-mix-aware planning, targeted investigation of high-LOS patterns, and repeatable evidence monitoring as new aggregate data becomes available.

STAGE 6 • STRATEGIC INSIGHTS & CLINICAL ADVISORY COMPLETE

Evidence-to-Action Matrix & 4 Priority Frameworks Synthesized

Advance to Stage 7 to export executive summaries, download raw datasets, and print the formal capstone project dossier.

Proceed to Stage 7: Reports & Export →

ED Analytics

DAMO-6994 Capstone

Analysis Pipeline

100%

?

About Project

Prep & Quality Engine

Dataset Explorer

Hypothesis Testing & St...

Executive Dashboard

Strategic Insights

Reports & Export

HA

Enterprise Workspace

CIHI NACRS Analytics

< Strategic Insights

Stage 7/7 · Reports & Export

EXPORT DOSSIER VALIDATED

STAGE 07 · DECISION SUPPORT & AUDIT

Report Generation & Decision Dossier

Configure, preview, and generate audit-ready healthcare executive dossiers, research reports, and clean data archives derived from CIHI NACRS emergency department analytics.

Grade AAA Accredited

ED Visits

7,296 Records

Quality: 98.5/100

REPORT CUSTOMIZATION

CUSTOM REPORT NAME

CIHI_NACRS_ED_Operational_Modeling_Dos

Letters, numbers, underscores, or hyphens only.

EXPORT FORMAT FILE

PDF (Portable Document Format)

SELECT REPORT CHAPTERS

1. Research Abstract

Introduction and problem context

2. Quality & Harmonization

Cohorts, record volume, validation index

3. Analytics Methodology

Hypothesis specifications & model definitions

4. AI Clinical Decisions

Recommendations & operational implications

5. Parametric Models

WLS regression, Mann-Kendall, SES forecast

QUICK DIRECT EXPORT

CSV

XLSX

ZIP

Generate PDF (choose "Save as PDF")

Direct Print Portfolio

Encrypted transmission. Report generation is compliant with PHIPA administrative standards.

Live WYSIWYG Document Preview

Selected chapters: 5 of 5

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EMERGENCY DEPARTMENT INTELLIGENCE PORTFOLIO

DOSSIER FILENAME: CIHI_NACRS_ED_Operational_Modeling_Dossier.pdf

1. RESEARCH ABSTRACT & EXECUTIVE SUMMARY

Emergency departments function as the primary safety net of the Canadian healthcare system. However, unprecedented patient volumes, rising clinical acuity, and systemic bed shortages have led to chronic ED crowding and prolonged Length of Stay (LOS). Operational leaders lack real-time predictive systems to forecast daily patient volumes, assess triage bottlenecks (particularly for CTAS Level 3 'Urgent' cases), and align nurse staffing rosters.

2. CLINICAL DATA QUALITY & COHORT HARMONIZATION

To facilitate statistical calibration, we compiled a longitudinal clinical cohort of **7,296 visit records**. The unified database completed strict schema matching, CTAS triage coding standardization, and missing duration interpolations, earning a validation compliance score of **98.5% (Grade AAA)**.

3. RESEARCH OBJECTIVES & METHODOLOGY

The core scientific framework maps five pre-specified hypotheses: (a) Evaluating median LOS variance across CTAS acuity tiers (Weighted Kruskal-Wallis, $\epsilon^2 = 0.7251$), (b) Testing admission status disparity (Weighted Mann-Whitney U, $r_b = 0.9981$), (c) Modeling stay duration determinants via Weighted Least Squares (WLS, $R^2 = 0.6256$), (d) Measuring life-stage demographic variations (Weighted Kruskal-Wallis, $\epsilon^2 = 0.7218$), and (e) Testing independence of Patient Sex and Visit Disposition (Pearson Chi-Square, $\chi^2 = 18,164.97$, Cramér's V = 0.0102). In addition, longitudinal trends are evaluated using the Mann-Kendall test (Z = 5.5977) and

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